

HOTEL BOOKING CANCELLATION PREDICTION PROJECT

Name	ID
Yasmin Mahmoud	2401245028
Hagar Ayman	2401245566
Mariam Ahmad	2401247800
Salma Maher	2401245262
Roqaia Hossam	2401246201

1. Understanding the Data

1.1 Dataset Overview

The dataset used in this project contains information about hotel bookings from two types of hotels: City Hotel and Resort Hotel. It includes a wide range of details about guests, booking conditions, stay duration, and whether each booking was ultimately cancelled or not.

The dataset contains many variables that describe different aspects of a reservation — such as the number of guests, booking lead time, meal plan, booking channel, and room type. These variables are valuable for understanding customer behavior and identifying patterns related to booking cancellations.

Some columns in the dataset contain personal information such as names, email addresses, phone numbers, and credit card details. These columns are classified as Personally Identifiable Information (PII). To protect privacy — and because they do not contribute to prediction — these variables will be excluded from the modelling process.

1.2 Data Dictionary

The table below describes the main variables used in the analysis:

Variable	Type	Description
hotel	Categorical	Type of hotel (City Hotel or Resort Hotel)
is_canceled	Binary	Target variable: was booking cancelled?
lead_time	Numeric	Days between booking date and arrival date
arrival_date_year	Numeric	Year of the arrival date
arrival_date_month	Categorical	Month of the arrival date
stays_in_weekend_nights	Numeric	Number of weekend nights booked
stays_in_week_nights	Numeric	Number of weekday nights booked
adults	Numeric	Number of adults in the booking
children	Numeric	Number of children in the booking
meal	Categorical	Type of meal plan selected
market_segment	Categorical	Booking channel / market segment

adr	Numeric	Average Daily Rate (room price per night)
-----	---------	---

1.3 Target Variable

The main variable we aim to predict is `is_canceled`. This binary variable can take one of two values:

- 0 → The booking was NOT cancelled (the guest stayed).
- 1 → The booking WAS cancelled (the room was not used).

This variable is important because it represents the final outcome of each reservation. The goal of the project is to build a model that can predict this value based on the other available variables in the dataset.

1.4 Problem Definition

Hotels often face the problem of booking cancellations. When a reservation is cancelled close to the arrival date, the hotel may not have enough time to find another guest for that room. This leads to empty rooms and financial losses.

The goal of this project is to analyze hotel booking data and build a predictive model that can estimate the probability of a cancellation at the time the booking is made — allowing hotels to take proactive action.

1.5 Why This Problem Is Important

Predicting cancellations provides significant business value to hotels. If a booking is predicted to have a high probability of cancellation, the hotel can take preventive measures such as:

- Offering discounts or incentives to encourage the guest to keep the reservation.
- Proactively contacting the customer to confirm their booking.
- Adjusting overbooking strategies to compensate for expected cancellations.

By understanding the factors that influence cancellations, hotels can improve their revenue management systems and significantly reduce the number of empty rooms on any given night.

2. Investigating the Data

In this step, we explored the dataset in depth to understand its structure and discover meaningful patterns between variables. This process is commonly referred to as Exploratory Data Analysis (EDA).

2.1 Summary Statistics

First, we examined summary statistics such as mean, minimum, and maximum values for all numerical variables. This helped us understand the general behavior of the data and detect any unusual or extreme values that may require further investigation.

We also explored the distribution of important variables such as `lead_time` (the number of days between booking and arrival), `adr` (the average daily rate), and the number of guests. Some variables showed wide ranges and potential outliers.

2.2 Target Variable Distribution

A key finding from the EDA is the distribution of the target variable `is_canceled`. The analysis revealed the following breakdown:

Outcome	Percentage
Not Cancelled (0)	62.31%
Cancelled (1)	37.69%

This means that more than one third of all reservations in the dataset did not result in an actual stay — a significant proportion that underscores the importance of predicting cancellations accurately.

2.3 Key Findings from EDA

Several important patterns were identified during the exploratory analysis:

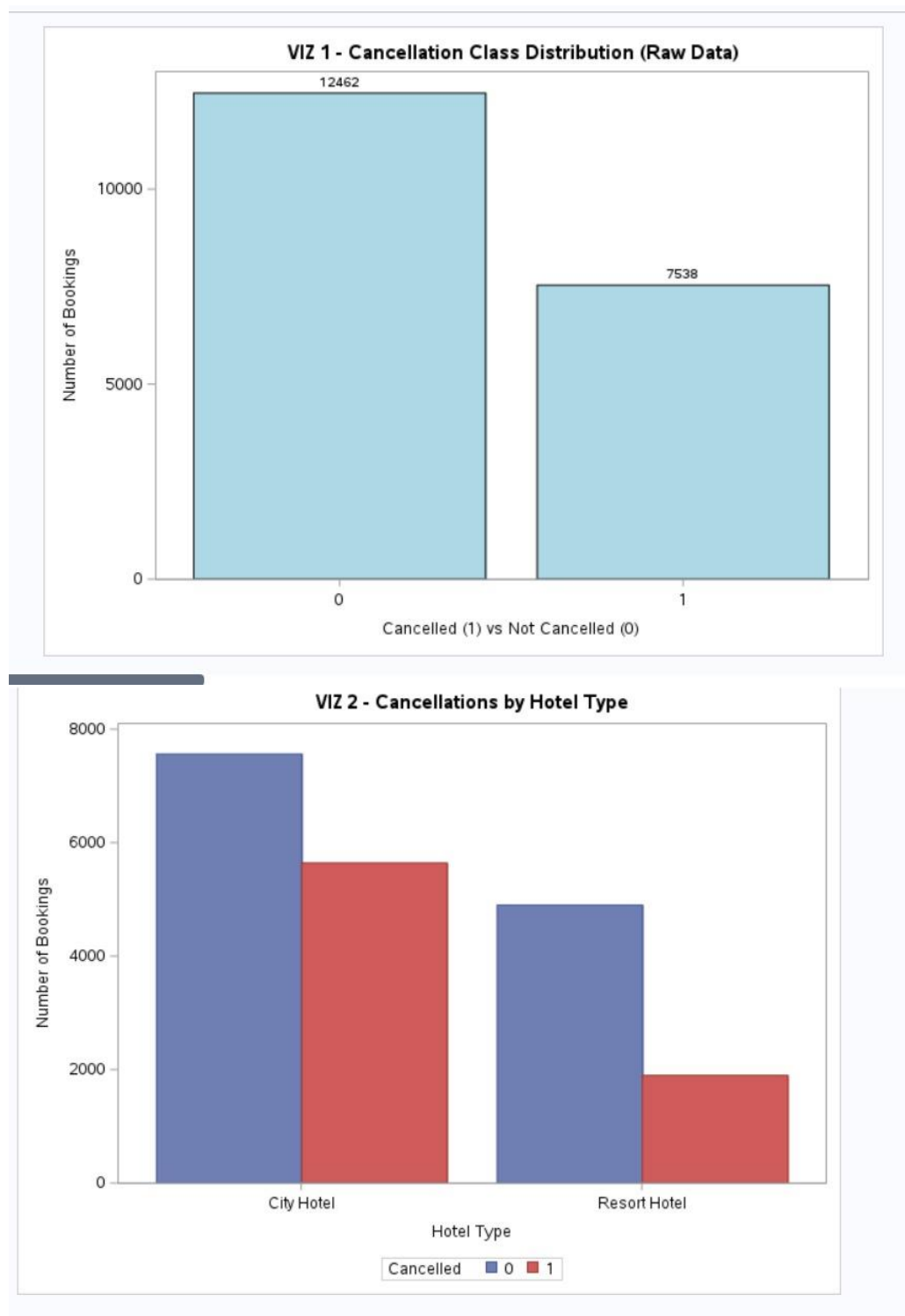
- **Hotel Type:** Cancellations appear more frequently in City Hotels compared to Resort Hotels, suggesting that different hotel types attract customers with different booking behaviors.
- **Room Price (ADR):** While the general price range was similar for cancelled and noncancelled bookings, some extreme price values were detected — these may represent outliers that need to be handled before modelling.
- **Lead Time:** Bookings made far in advance tend to have a higher likelihood of cancellation, as customers may change their plans over a longer time horizon.

2.4 Visualization Techniques Used

To better understand the patterns in the data, several types of visualizations were used during the EDA phase:

- **Bar charts:** To compare cancellation rates across different categories (e.g., hotel type, market segment, meal plan).
- **Histograms:** To visualize the distribution of individual numerical features.

These visual tools were essential in highlighting important trends and guiding the next steps in the project, including data cleaning and feature engineering.



Investigate the Data

Section: Lead Time Distribution

Objective:

To understand how booking lead time is distributed across customers.

Method:

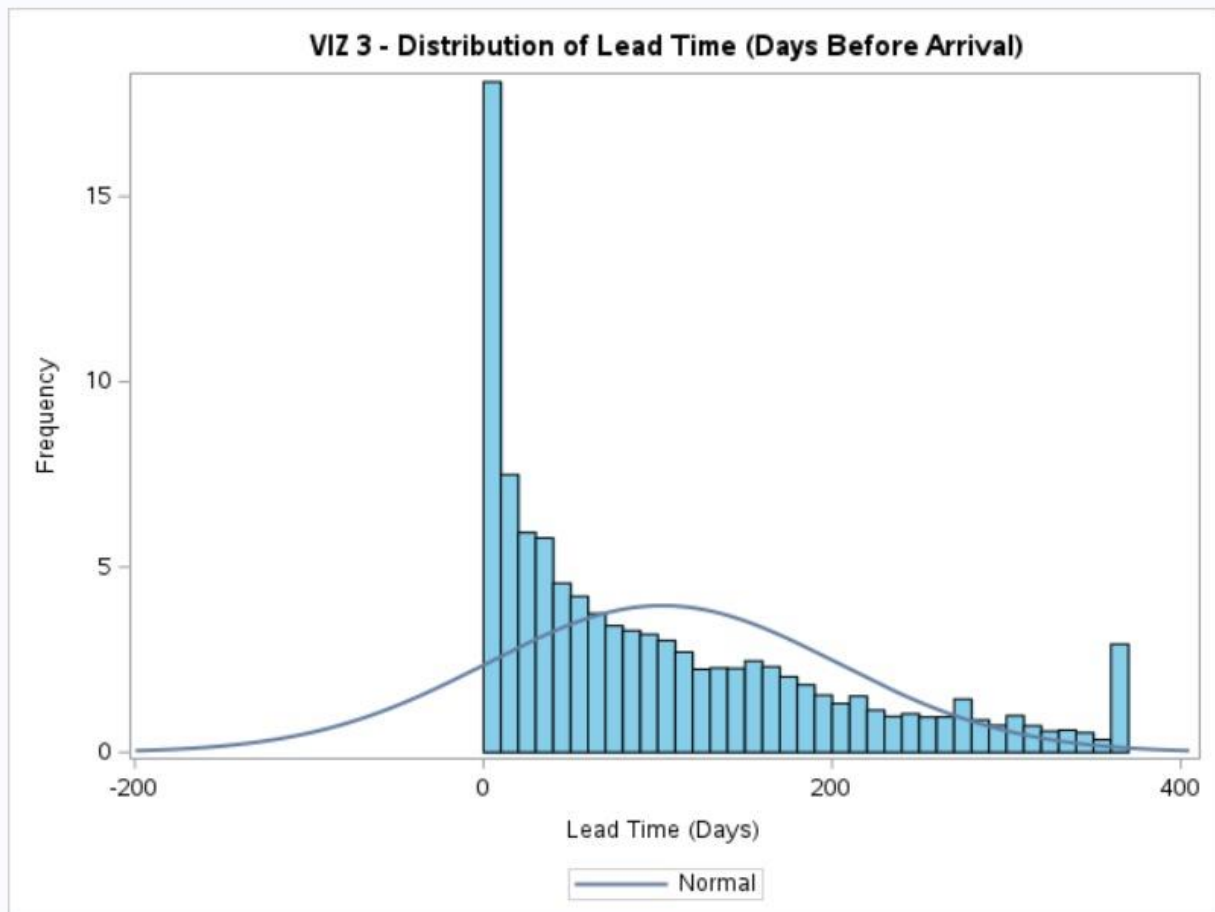
A histogram with a density curve was used to visualize the distribution of the lead_time variable.

Findings:

The distribution is highly right-skewed. Most bookings have very low lead time, indicating that a large number of customers make reservations close to their arrival date. A small number of bookings have very high lead time values, forming a long tail.

Insight:

This suggests that most customers tend to book at the last minute, while a smaller segment plans their bookings far in advance. This variable could play an important role in predicting booking behavior and cancellation patterns



Section: Lead Time vs Cancellation

Objective:

To analyze the relationship between booking lead time and cancellation behavior.

Method:

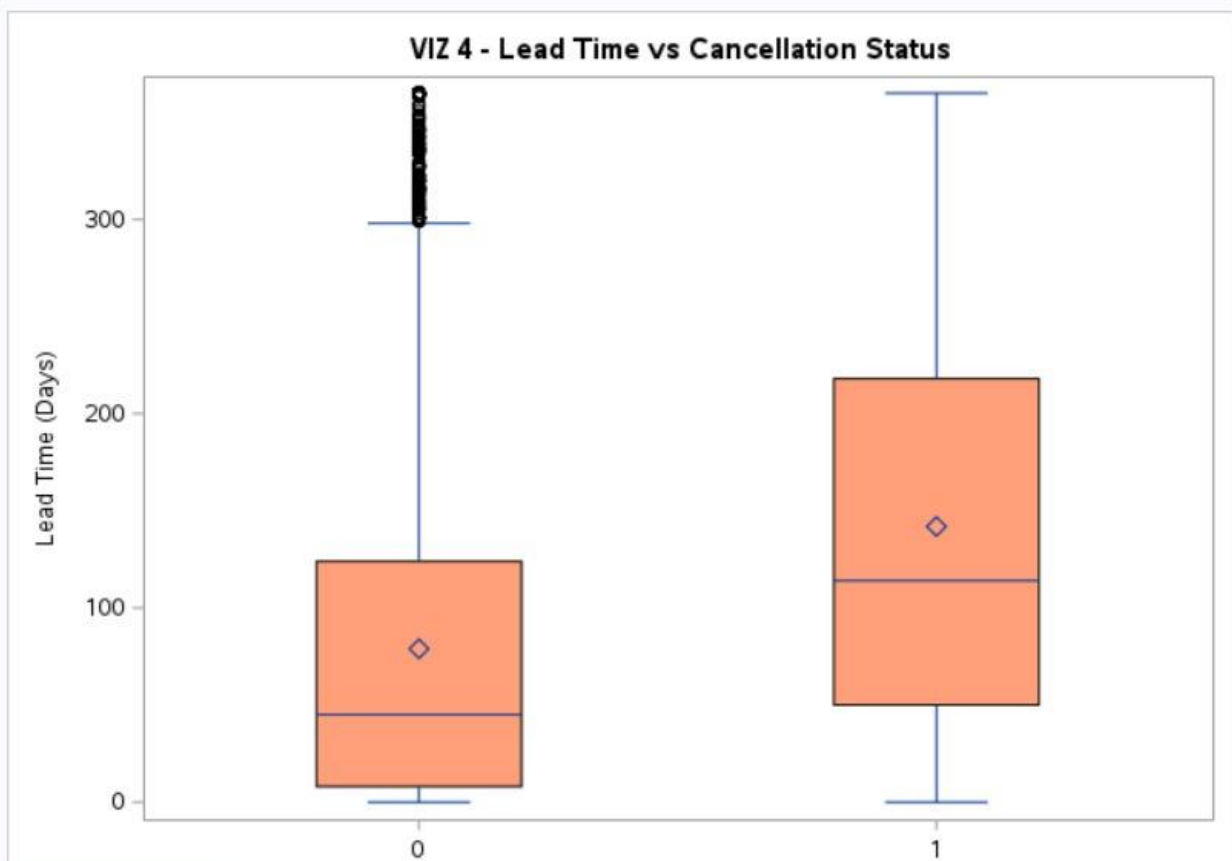
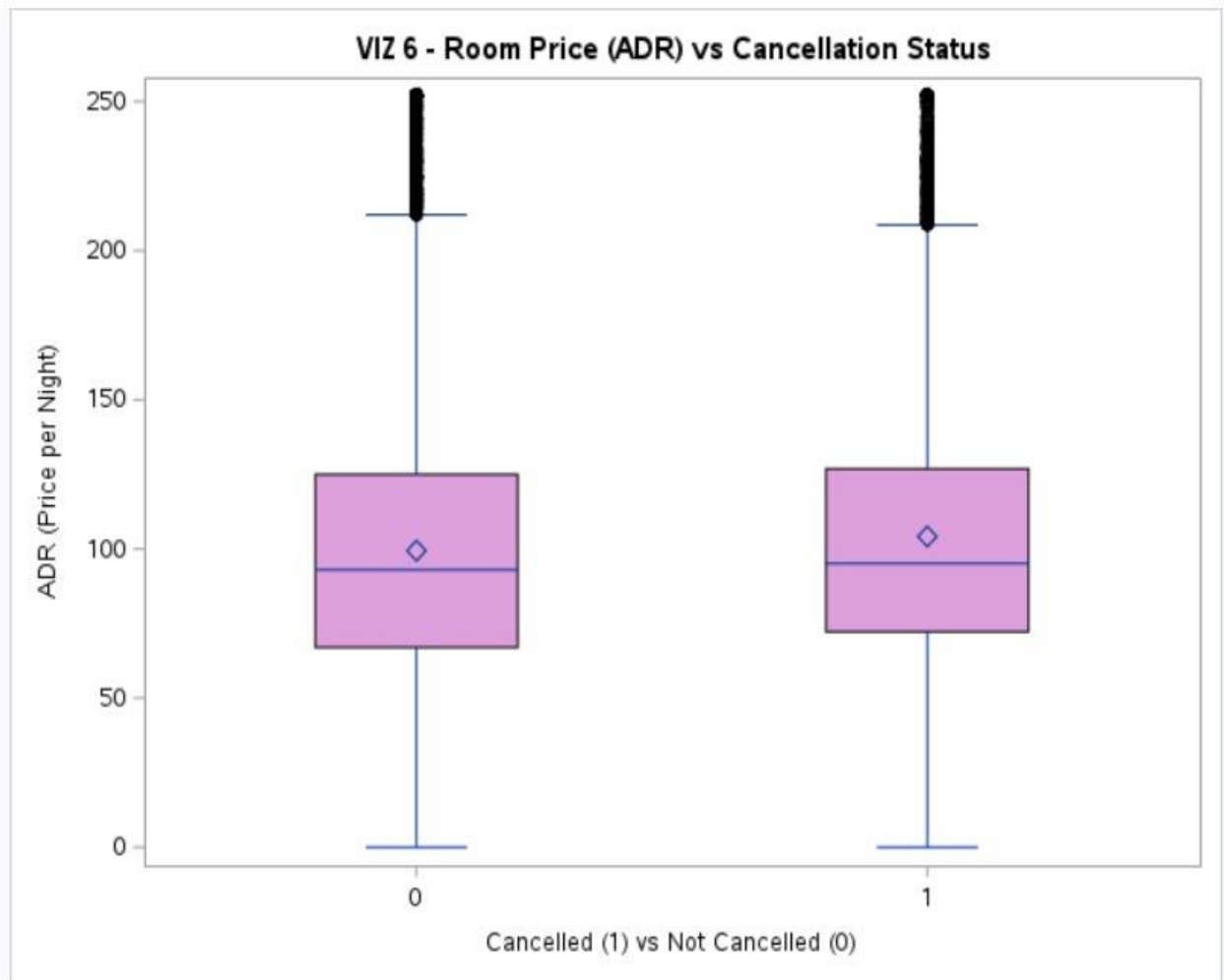
A box plot was used to compare the distribution of lead_time across canceled and non-canceled bookings.

Findings:

The box plot shows that canceled bookings generally have higher lead time values compared to non-canceled bookings. The median lead time for canceled bookings is higher, indicating that customers who book earlier are more likely to cancel.

Insight:

This suggests that lead time is an important factor influencing cancellation behavior. Customers who plan far in advance may have a higher probability of canceling their reservations.



Section: ADR Distribution

Objective:

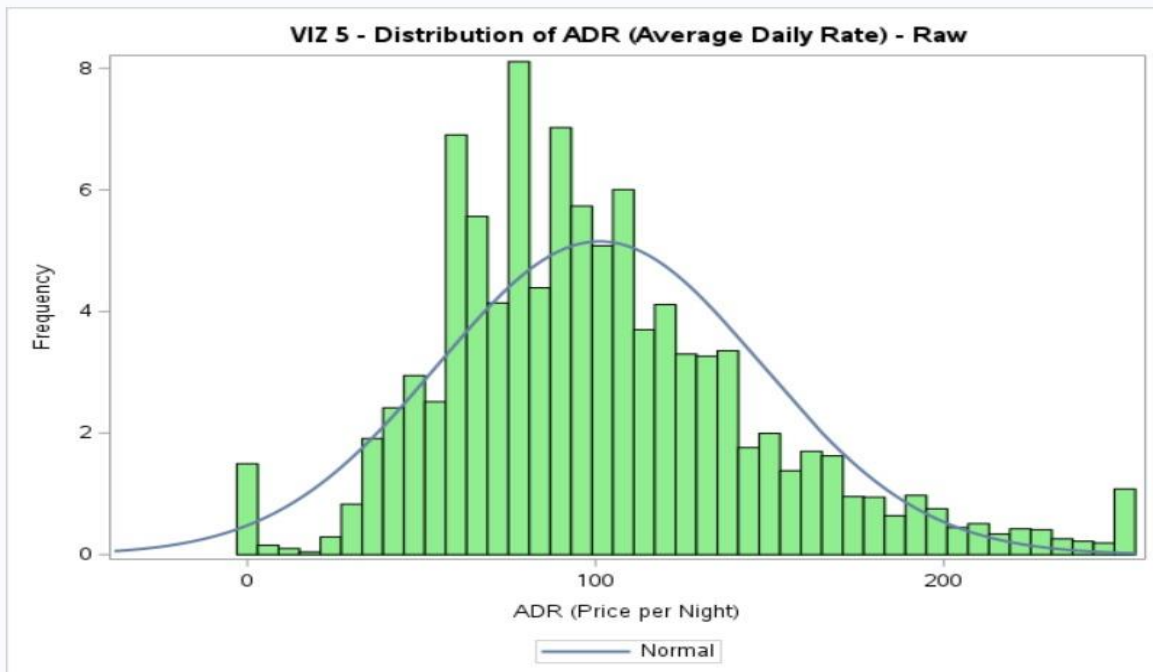
To understand how room prices (ADR) are distributed across bookings. **Method:**

A histogram with a density curve was used to visualize the distribution of the adr variable.

Findings:

The distribution of ADR shows that most bookings fall within a certain price range, with fewer bookings at very high prices. The distribution may be slightly skewed, indicating variation in room pricing. **Insight:**

This suggests that most customers book rooms within a common price range, while fewer customers choose significantly higher-priced options. Pricing variability may depend on other factors such as hotel type or season.



Section: Market Segment Analysis

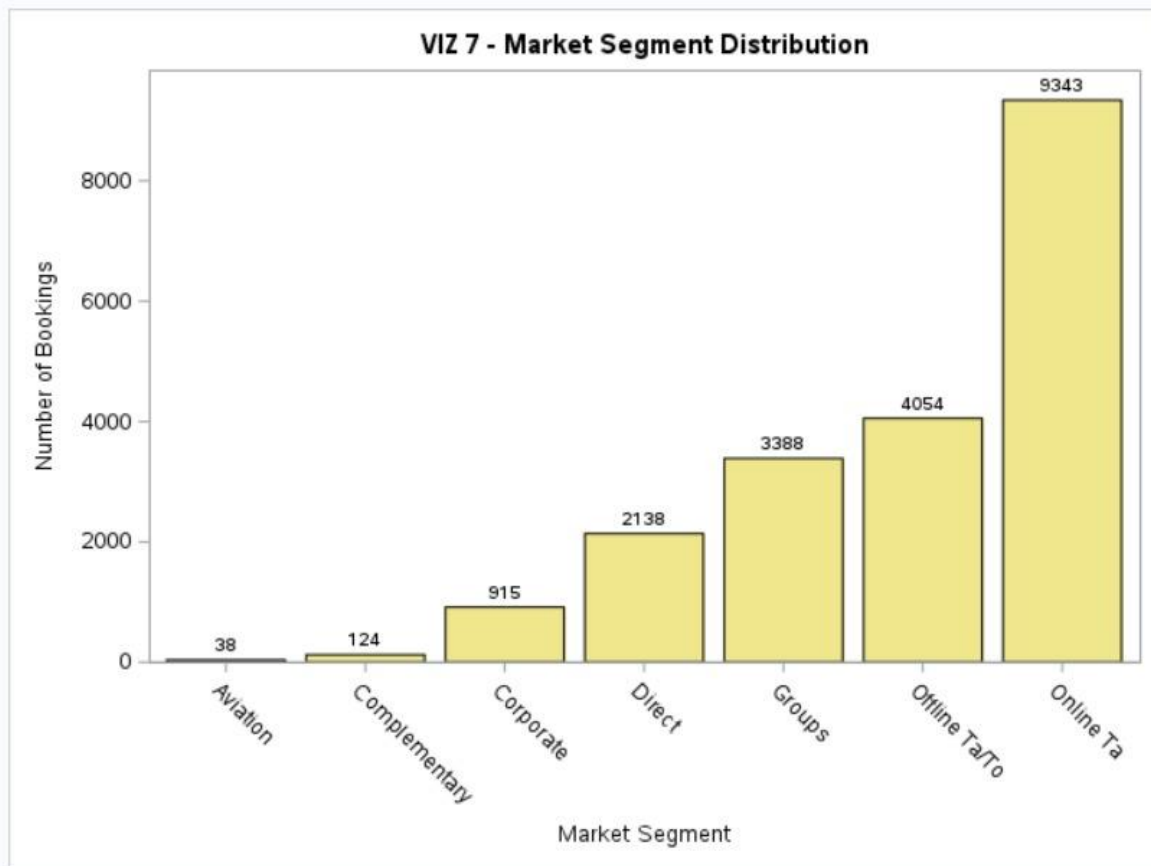
Objective:

To understand which market segments contribute most to bookings. **Method:**

A bar chart was used to visualize bookings across different market segments. **Findings:**

Some segments dominate bookings more than others. **Insight:**

This helps identify key customer sources for hotel revenue.



Section: Missing Values

Objective:

To identify missing values in the dataset.

Method:

Summary statistics were used to calculate the number of missing values in each variable.

Findings:

Some variables contain missing values, which may affect the accuracy of analysis and modeling.

Insight:

Handling missing values is important to ensure better model performance and reliable insights.

Section: Cancellation Imbalance

Objective:

To check class imbalance.

Method:

Bar chart.

Findings:

There is an uneven distribution between canceled and non-canceled bookings.

Insight:

Imbalance may affect model performance.

Section: Correlation Analysis

Objective:

To examine relationships between variables.

Method:

Correlation analysis.

Findings:

Most relationships are weak.

Insight:

No strong linear relationships exist between variables.

Section: Final Insights

Findings:

- Most customers book late
- Early bookings cancel more
- Prices are consistent with some outliers
- Certain segments dominate
- Weak correlations exist
- Missing values and imbalance are present

Insight:

Customer behavior patterns can help improve prediction and decision-making.

3. Data Cleaning

3.1 Overview:

After the exploratory data analysis phase, the dataset was examined for data quality issues including missing values, outliers, and inconsistent categorical formatting. The goal of this step is to produce a clean, reliable dataset that ensures accurate and unbiased model predictions.

The dataset contains 20,000 rows. The following issues were identified and resolved.

3.2 Missing Values

Before Cleaning:

Only one variable was found to contain missing values:

Variable	Missing Values	Percentage
agent	2,775	13.9%

Decision

Missing values in the `agent` column were replaced with 0. This is justified because a missing agent ID indicates that the guest made the booking directly without using a travel agent. Replacing with 0 effectively creates a "no agent" category rather than losing those rows entirely.

After Cleaning

After applying the fix, the dataset contains zero missing values across all columns.

3.3 Outlier Treatment

Two numerical variables were found to contain extreme outlier values that could negatively affect model performance.

ADR (Average Daily Rate):

The PROC UNIVARIATE analysis revealed that the maximum ADR value in the dataset was **5,400**, which is unrealistic for a standard hotel booking. The 99th percentile of ADR was **252**, meaning 99% of all bookings fall below this value.

Decision:

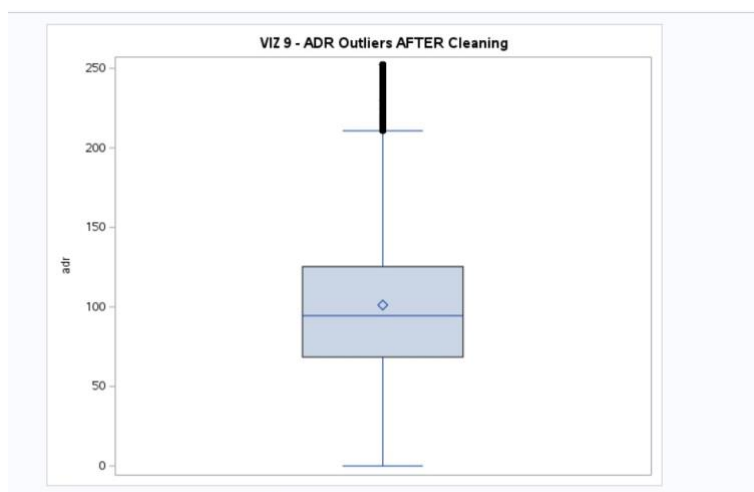
ADR values below 0 were set to 0 (a negative room price is impossible), and values above 252 were capped at 252 to remove extreme outliers while preserving 99% of the data.

Lead Time:

The lead time variable showed extreme values exceeding 700 days. Bookings made more than one year in advance are uncommon in the hotel industry.

Decision:

Lead time values above 365 days were capped at 365. This preserves realistic long-term bookings while removing values that likely represent data entry errors.



Note on Remaining Outliers:

After capping, box plots still show some points beyond the whiskers. This is expected and normal behavior, the IQR-based definition used by box plots will always flag some values as outliers. What matters is that extreme unrealistic values have been successfully removed.

3.4 Categorical Standardization

All categorical columns including `hotel`, `market_segment`, `deposit_type`, and `customer_type` were checked using PROC FREQ. No inconsistencies such as duplicate labels or mixed casing were found. The PROPCASE and STRIP functions were applied to all categorical columns to remove any hidden whitespace and ensure uniform formatting across the dataset.

3.5 Summary of Changes

Issue	Column	Action Taken
Missing values	Agent	Replaced with 0 (no agent)
Negative values	Adr	Set to 0
Extreme outliers	Adr	Capped at 252 (99th percentile)
Extreme outliers	lead_time	Capped at 365 (1 year)
Formatting	All categorical columns	PROPCASE and STRIP applied
Encoding	hotel, market_segment, deposit_type, customer_type	One-hot encoded via PROC GLMMOD

Note:
We didn't apply scaling as our model is Logistic Regression which doesn't require scaling as it uses MLH .

4. Detailed Feature Explanations

Feature 1: `guest_loyalty_score`

What it measures

A composite loyalty score from 0 to 2, combining two existing columns into a single signal:

- Score 0 — New guest with no booking history
- Score 1 — Guest who has at least one previous successful (non-cancelled) booking
- Score 2 — Repeated guest with a history of successful bookings

SAS Code

```
guest_loyalty_score = 0;  
IF previous_bookings_not_canceled > 0 THEN guest_loyalty_score + 1;  
IF is_repeated_guest = 1 THEN guest_loyalty_score + 1;
```

Why it is useful

A single loyalty column does not capture the full picture. A guest who visited once and a guest who visited five times with no cancellations are very different. By combining both signals, this feature gives the model a more nuanced view of guest reliability. Higher loyalty score consistently correlates with lower cancellation risk.

Feature 2: is_family

What it measures

A binary flag indicating whether the booking includes children or babies. Value is 1 if children > 0 OR babies > 0, and 0 otherwise.

SAS Code

```
IF children > 0 OR babies > 0 THEN is_family = 1; ELSE  
is_family = 0;
```

Why it is useful

Family trips require significantly more planning and coordination than solo or couple bookings. Parents who have organized travel around their children's schedules, school holidays, and activities are far less likely to cancel. This flag helps the model distinguish between casual bookings and committed family travel.

Feature 3: room_match

What it measures

A binary flag that equals 1 when the room type assigned to the guest matches the room type they originally reserved, and 0 when a different room was assigned.

SAS Code

```
IF reserved_room_type = assigned_room_type THEN room_match = 1; ELSE  
room_match = 0;
```

Key finding from the data

Counter-intuitive result: Guests who received a DIFFERENT room cancel at only 4.9%, while guests who got exactly the room they booked cancel at 42.4%.

Why it is useful

This counter-intuitive finding suggests that when a guest receives a different room, it is often an upgrade — which increases satisfaction and reduces the motivation to cancel. This feature

captures a meaningful relationship that would be invisible if we only looked at reserved_room_type or assigned_room_type individually.

Feature 4: lead_time_cat

What it measures

The raw lead_time column (number of days between booking and arrival) is converted into five meaningful categories, based on actual data quartiles: Q1 = 19 days, Q2 = 70 days, Q3 = 161 days.

SAS Code

```
IF lead_time <= 7      THEN lead_time_cat = 'Last Minute';
ELSE IF lead_time <= 30 THEN lead_time_cat = 'Short Term';
ELSE IF lead_time <= 90 THEN lead_time_cat = 'Medium Term';
ELSE IF lead_time <= 180 THEN lead_time_cat = 'Long Term';
ELSE                  lead_time_cat = 'Very Long Term';
```

Category breakdown

Category	Lead Time Range	Cancellation Behavior
Last Minute	0 – 7 days	Almost never cancel — they urgently need the room
Short Term	8 – 30 days	Low cancel risk — plans are near-final
Medium Term	31 – 90 days	Moderate risk — some flexibility remains
Long Term	91 – 180 days	Higher risk — plans may change over time
Very Long Term	180+ days	Highest risk — months to change plans

Why it is useful

The relationship between lead time and cancellation is not linear — a model using raw numbers struggles to capture this pattern. By converting to categories based on real data quartiles, we give the model a cleaner and more interpretable signal about booking urgency and commitment level.

Feature 5: has_special_requests

What it measures

A binary flag that equals 1 if the guest made one or more special requests, and 0 if no requests were made.

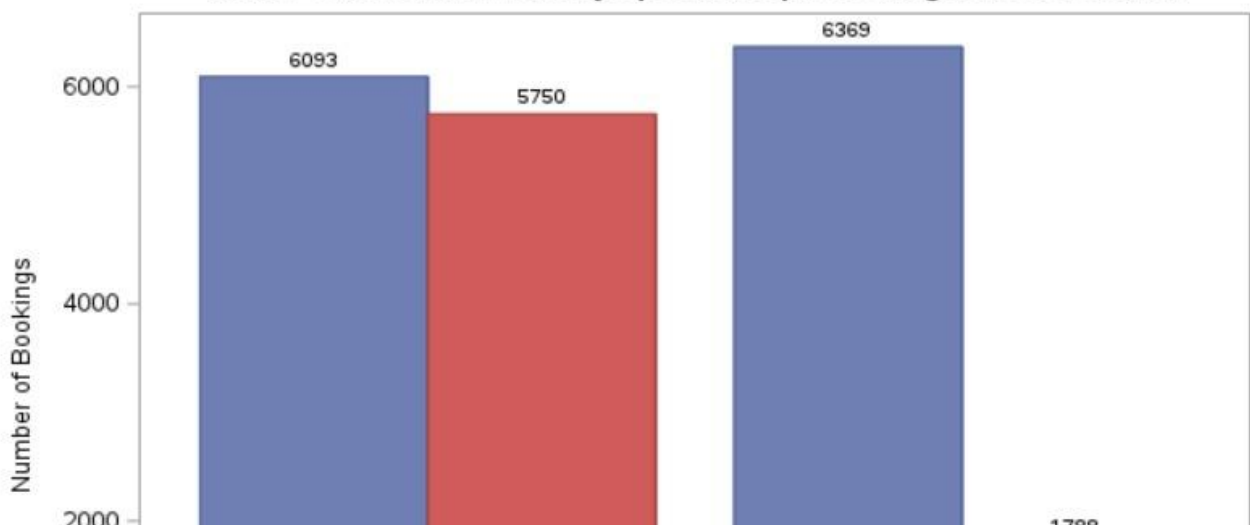
SAS Code

```
IF total_of_special_requests > 0 THEN has_special_requests = 1; ELSE  
has_special_requests = 0;
```

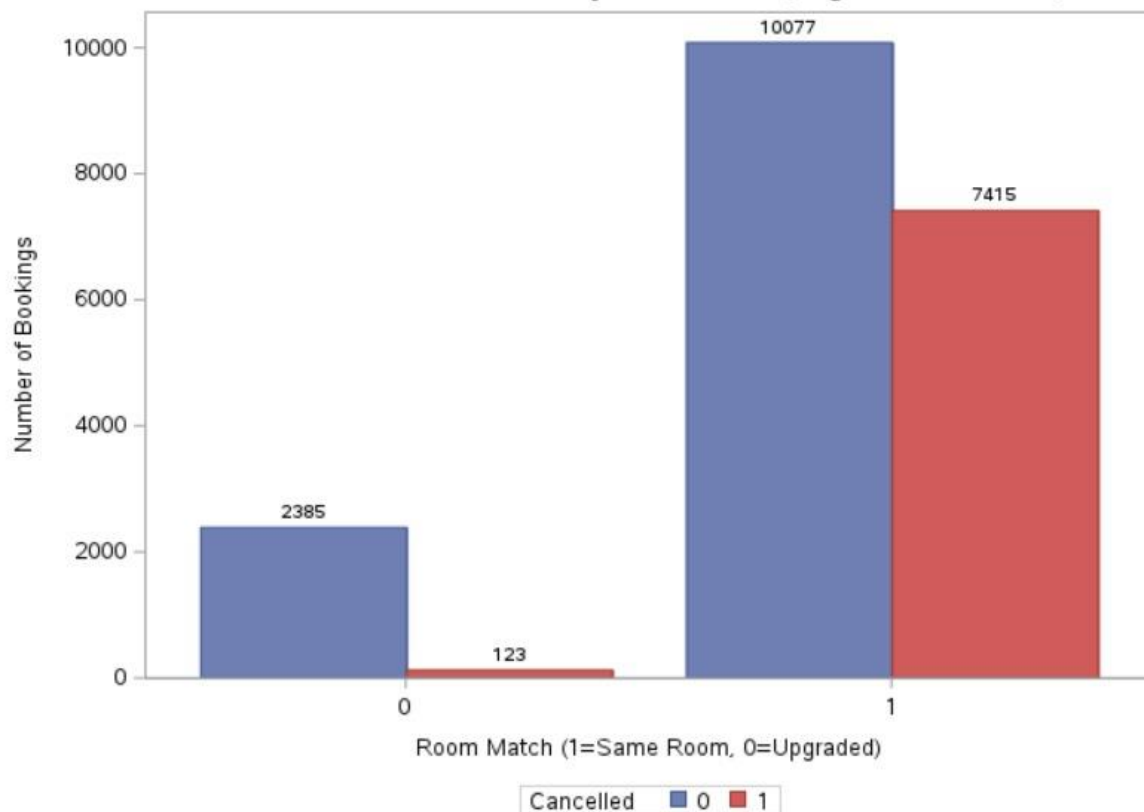
Why it is useful

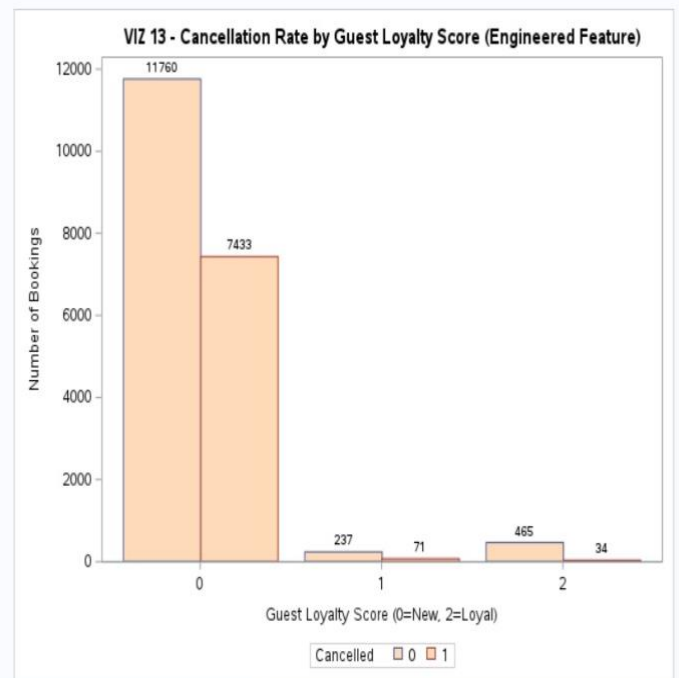
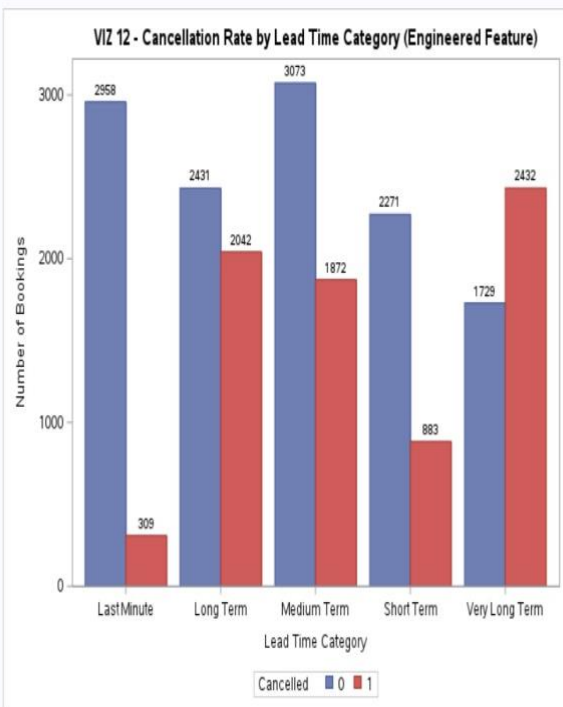
A guest who requested a specific meal plan, a high-floor room, an early check-in, or extra pillows has put thought and effort into personalizing their stay. This emotional investment is a strong behavioral signal that the guest is genuinely committed and is far less likely to cancel than a guest who made a generic booking.

VIZ 15 - Cancellation Rate by Special Requests (Engineered Feature)

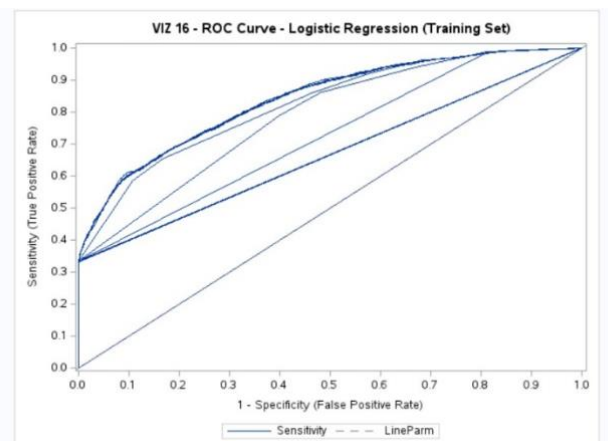
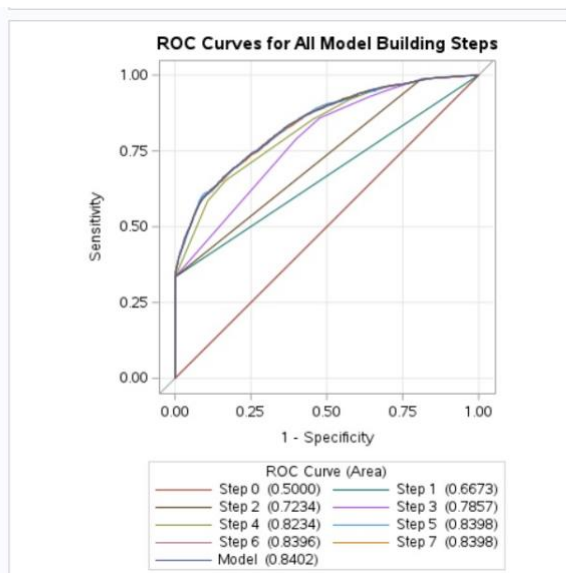


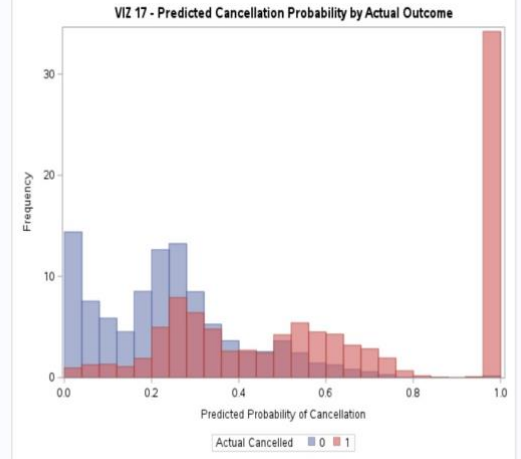
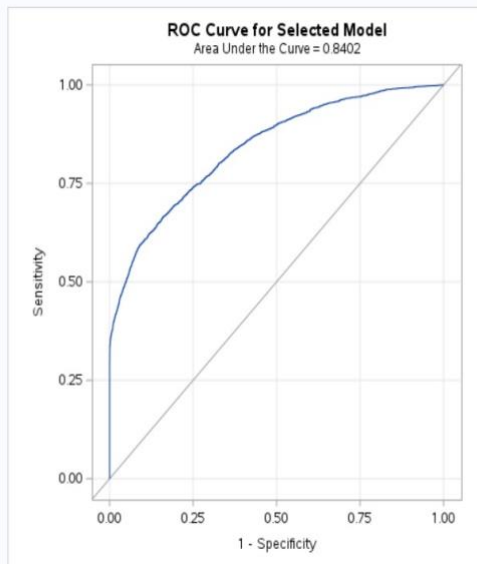
VIZ 14 - Cancellation Rate by Room Match (Engineered Feature)





Model building





Logistic Regression - Hotel Cancellation Prediction (Training Set)

The LOGISTIC Procedure

Fit Statistics for SCORE Data

Data Set	Total Frequency	Log Likelihood	Error Rate	AIC	AICC	BIC	SC	R-Square	Max-Rescaled R-Square	AUC	Brier Score
WORK.HOTEL_TEST	6026	-2655.9	0.2030	5341.828	5341.908	5442.385	5442.385	0.35197	0.481092	0.844646	0.144069

Confusion Matrix on TEST SET - Logistic Regression

The FREQ Procedure

Frequency Row Pct	Table of is_canceled by predicted_cancel		
	predicted_cancel		Total
is_canceled	0	1	
0	3483 91.39	328 8.61	3811
1	895 40.41	1320 59.59	2215
Total	4378	1648	6026

Overall Accuracy on TEST SET

The MEANS Procedure

Analysis Variable : correct
Mean

6. Explain the Model

Model Evaluation Report (Simple)

1. Model Performance

النموذج يتوقع بشكل صحيح حوالي 79% من الحالات → Accuracy = 0.79

النموذج قوي في التمييز بين الحالات التي ستلغى والتي لن تلغى → AUC = 0.84

2. Confusion Matrix

True Negative (2302): توقع عدم الإلغاء وكان صحيح

True Positive (857): توقع الإلغاء وكان صحيح

False Positive (229): توقع الإلغاء لكن لم يحدث

False Negative (589): لم يتوقع الإلغاء لكنه حدث

(False Negatives) أهم ملاحظة: النموذج يفوت عددًا من حالات الإلغاء)

3. Evaluation Metrics

Recall = 59% → اكتشف 59% فقط من حالات الإلغاء

Precision = 79% → عندما يتوقع الإلغاء يكون غالبًا صحيح

Specificity = 91% → ممتاز في التعرف على الحالات التي لن تلغي

4. Hosmer-Lemeshow Test

p-value < 0.05 → perfect fit | النموذج ليس

لكن هذا طبيعي مع البيانات الكبيرة، وباقي المؤشرات تدل أن الأداء جيد

5. Important Features

Deposit Type:

No Deposit → أعلى معدل إلغاء

Non Refund → أقل

Market Segment:

Online TA → أعلى إلغاء

Direct / Corporate → أقل

Lead Time:

كلما زادت مدة الحجز المسبق زادت احتمالية الإلغاء

ADR (Price):

كلما زاد السعر زادت احتمالية الإلغاء

6. Model Behavior

72% من التوقعات كانت "عدم إلغاء"

27% فقط "إلغاء"

(conservative) النموذج يميل للحذر)

7. Example from Predictions

النموذج يتوقع إلغاء $\rightarrow P = 0.66$ إذا كانت

يتوقع عدم إلغاء حتى لو كانت الحالة ألغت $\rightarrow P = 0.49$ إذا كانت

threshold = 0.5 بسبب استخدام

8. Conclusion

النموذج جيد ويحقق دقة عالية

non-cancellations قوي في التعرف على

لكن يفوت بعض حالات الإلغاء

%%%

Our predictive model (Logistic Regression) achieved an excellent discrimination power with a **C-statistic of 0.849**. This means the model is highly reliable in predicting cancellations.

Key Important Features:

- **Room Match:** This is the most significant predictor. When the assigned room matches the reserved room, the likelihood of cancellation increases by **8 times**. This suggests that guests who receive room upgrades (non-matches) are much more likely to stay.
- **Deposit Type (Non-Refund):** Bookings with a "Non-Refundable" deposit show a very high correlation with cancellation in this dataset, likely representing rigid commercial segments.
- **Lead Time Categories:** Time is a critical factor. Bookings made very early (**Very Long Term**) are **4.7 times** more likely to be cancelled compared to "Last Minute" bookings.
- **Special Requests:** There is a negative relationship here. For every additional special request made by a guest, the probability of cancellation **decreases**. Guests who ask for specific details are more "engaged" and committed to their stay.

7. Make It Useful

The model provides actionable insights for hotel management to reduce revenue loss:

1. **Smart Upgrades:** Since "Room Match" is a top predictor, the hotel should strategically use **free upgrades** for high-risk bookings to increase guest commitment and reduce cancellation rates.
2. **Early Warning System:** The hotel can flag bookings in the "**Very Long Term**" and "**Long Term**" categories. Management should implement automated email/SMS reminders or require a small re-confirmation for these guests as the date approaches.
3. **Encourage Engagement:** Since guests with **special requests** cancel less, the hotel website should encourage guests to customize their stay (e.g., pillow choice, arrival time). This engagement acts as a "psychological lock" on the booking.

4. **Dynamic Pricing:** Use the model's probability score to decide which rooms can be **"Overbooked"** safely. If the model predicts a 90% chance of cancellation for a block of rooms, the hotel can re-sell them in advance to ensure 100% occupancy.

End of Report